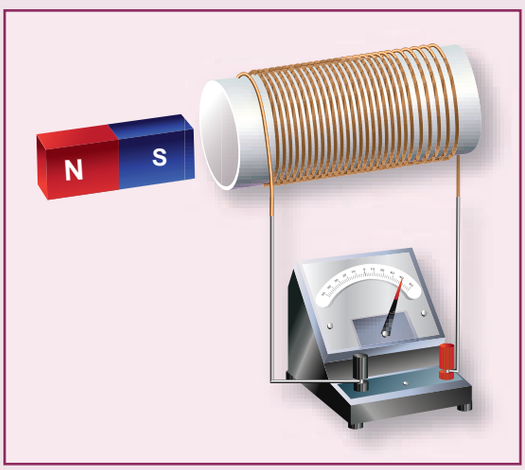
### Importance of Electromagnetic Induction!

The application of the phenomenon of Electromagnetic Induction is almost everywhere in the present day life. Right from home appliances to huge factory machineries, from cellphone to computers and internet, from electric guitar to satellite communication, all need electricity for their operation. There is an ever growing demand for electric power.

All these are met with the help of electric generators and transformers which function on electromagnetic induction. The modern, sophisticated human life would not be possible without the discovery of electromagnetic induction.

**ACTIVITY**

### Exploring Electromagnetic Induction




Make a circuit containing a coil of insulated wire wound around soft hollow core and a galvanometer as shown in Figure. It is better to use a thin wire for the coil so that we can wind many turns in the available space. Perform the steps described in first experiment of Faraday with the help of a strong bar magnet. Students will get hands-on experience about electromagnetic induction.

## EXAMPLE 4.3

A cylindrical bar magnet is kept along the axis of a circular solenoid. If the magnet is rotated about its axis, find out whether an electric current is induced in the coil.

### *Solution*

The magnetic field of a cylindrical magnet is symmetrical about its axis. As the magnet is rotated along the axis of the solenoid, there is no induced current in the solenoid because the flux linked with the solenoid does not change due to the rotation of the magnet.

## EXAMPLE 4.4

A closed coil of 40 turns and of area 200 cm$^2$, is rotated in a magnetic field of flux density 2 Wb m$^{-2}$. It rotates from a position where its plane makes an angle of 30° with the field to a position perpendicular to the field in a time 0.2 s. Find the magnitude of the emf induced in the coil due to its rotation.

### *Solution*

$N = 40$ turns; $B = 2$ Wb m$^{-2}$

$A = 200$ cm$^2$ $= 200 \times 10^{-4}$ m$^2$;

Initial flux, $\Phi_i = BA\cos\theta$

$= 2 \times 200 \times 10^{-4} \times \cos 60^\circ$

since $\theta = 90^\circ - 30^\circ = 60^\circ$

$\Phi_i = 2 \times 10^{-2}$ Wb

Final flux, $\Phi_f = BA\cos\theta$

$= 2 \times 200 \times 10^{-4} \times \cos 0^\circ$

since $\theta = 0^\circ$

$\Phi_f = 4 \times 10^{-2}$ Wb

The magnitude of the induced emf is

$$\varepsilon = N\frac{d\Phi_B}{dt}$$

$$= \frac{40 \times \left(4 \times 10^{-2} - 2 \times 10^{-2}\right)}{0.2} = 4\text{V}$$



UNIT-4(XII-Physics_Vol-1).indd 201 17-02-2020 13:17:40